

NERCHINSKIY ZAVOD, Russian Federation

WMO: 308790

Lat: 51.317N

Lon: 119.617E

Elev: 2031

StdP: 13.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.3	-30.2	-37.8	0.7	-33.7	-33.9	0.9	-29.7	11.0	-1.6	9.5	-2.2	0.0	90	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	85.1	64.7	81.5	63.9	78.2	62.9	68.8	78.9	67.0	76.2	65.3	74.1	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	101.6	73.5	63.8	95.4	71.4	62.0	89.6	69.7	34.4	79.2	32.8	76.5	31.4	73.9	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.0	9.4	-37.3	91.0	5.9	4.1	-41.5	94.0	-45.0	96.3	-48.3	98.6	-52.5	101.6		
(5)			-37.4	72.0	5.6	2.5	-41.4	73.8	-44.7	75.2	-47.9	76.6	-52.0	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.1	-15.4	-5.4	14.6	35.5	49.8	62.1	66.6	62.1	49.4	32.2	7.2	-11.6
	DBStd	30.13	8.71	9.80	11.49	8.55	7.23	6.43	4.69	5.85	7.46	9.04	11.97	9.16
	HDD50	9065	2028	1551	1098	441	93	3	0	3	99	554	1285	1910
	HDD65	13266	2493	1971	1563	885	473	126	35	123	469	1018	1735	2375
	CDD50	1437	0	0	0	7	88	366	515	378	82	1	0	0
	CDD65	162	0	0	0	0	2	39	86	34	1	0	0	0
	CDH74	1902	0	0	0	4	85	612	832	339	30	0	0	0
	CDH80	498	0	0	0	0	17	180	226	71	4	0	0	0
Wind	WSAvg	2.8	1.1	1.7	3.2	4.6	4.7	3.3	2.9	2.9	3.4	3.3	1.9	1.1
Precipitation	PrecAvg	17.2	0.1	0.2	0.3	0.8	1.1	2.7	4.8	3.9	2.1	0.6	0.4	0.2
	PrecMax	25.5	0.4	0.4	0.7	1.7	3.0	7.2	9.3	8.7	5.3	1.8	1.0	0.5
	PrecMin	11.1	0.0	0.0	0.0	0.1	0.2	0.4	1.3	0.9	0.2	0.1	0.0	0.0
	PrecStd	3.6	0.1	0.1	0.2	0.4	0.7	1.7	2.2	1.8	1.4	0.4	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	27.3	50.2	70.0	82.4	91.3	90.5	86.9	77.8	62.8	40.4	15.3
		MCWB	9.3	22.5	38.3	49.3	59.1	64.6	67.4	67.3	57.5	48.1	33.4	13.4
	2%	DB	5.0	19.2	41.6	62.5	75.5	84.6	85.5	81.3	72.3	55.4	33.2	9.7
		MCWB	3.7	16.4	33.0	45.8	55.8	62.7	66.0	64.6	56.8	43.7	28.2	8.5
	5%	DB	0.8	14.2	36.4	56.3	70.4	80.5	81.8	77.5	67.8	50.7	27.9	4.9
		MCWB	-0.2	12.2	29.5	42.3	53.3	61.9	65.6	63.3	54.6	40.8	24.5	3.9
	10%	DB	-3.0	9.6	31.9	51.0	65.7	76.6	78.4	74.1	63.3	46.1	24.0	1.4
		MCWB	-3.8	8.0	26.8	39.1	50.4	60.9	64.6	62.0	53.0	38.2	21.4	0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	22.4	39.2	51.5	61.6	69.0	72.6	70.0	62.2	48.5	33.8	13.4
		MCDB	10.9	27.5	49.1	68.5	78.3	83.1	85.2	80.5	69.2	60.5	39.8	15.0
	2%	WB	3.8	16.6	33.5	46.7	57.8	66.4	69.7	67.4	59.0	44.8	28.7	8.6
		MCDB	5.0	19.3	41.2	60.9	72.4	78.9	80.1	76.5	66.9	53.9	32.9	9.9
	5%	WB	-0.1	12.3	29.9	42.9	54.3	64.1	67.9	65.6	56.5	41.7	24.7	4.0
		MCDB	0.8	14.2	35.9	56.0	68.1	76.0	76.7	74.3	65.7	49.6	27.7	4.9
	10%	WB	-3.8	8.1	26.9	39.5	51.6	62.2	66.3	63.6	54.1	38.7	21.3	0.7
		MCDB	-3.1	9.6	31.7	49.9	64.1	73.4	74.7	71.5	62.4	45.5	23.9	1.5
Mean Daily Temperature Range	5% DB	MDBR	14.8	19.3	21.0	20.9	23.4	23.6	20.4	20.8	22.3	19.7	17.6	14.1
		MCDBR	17.8	22.3	24.4	30.9	33.5	31.3	27.9	26.8	29.1	26.9	20.0	16.0
		MCWBR	16.5	19.8	16.7	18.5	17.3	13.5	11.7	12.2	14.6	17.0	15.6	14.8
	5% WB	MCDBR	17.7	22.4	23.2	28.5	29.9	26.1	22.7	23.1	24.5	25.7	19.5	15.8
		MCWBR	16.4	19.9	16.8	18.0	16.5	13.1	11.6	12.1	13.0	16.8	15.4	14.7
Clear-Sky Solar Irradiance	taub	0.249	0.273	0.324	0.421	0.480	0.453	0.419	0.390	0.346	0.317	0.276	0.257	
	taud	2.218	2.183	2.107	2.006	1.925	2.121	2.280	2.334	2.396	2.369	2.275	2.161	
	Ebn at Noon	246	270	274	257	246	256	264	266	266	252	231	217	
	Edh at Noon	22	30	40	50	57	47	40	36	30	25	20	19	
All-Sky Solar Radiation	RadAvg	484	849	1351	1645	1802	1902	1740	1557	1213	817	492	370	
	RadStd	13	29	61	92	142	180	122	130	91	60	21	17	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (6 neighbors)		N/A	N/A	-3.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page